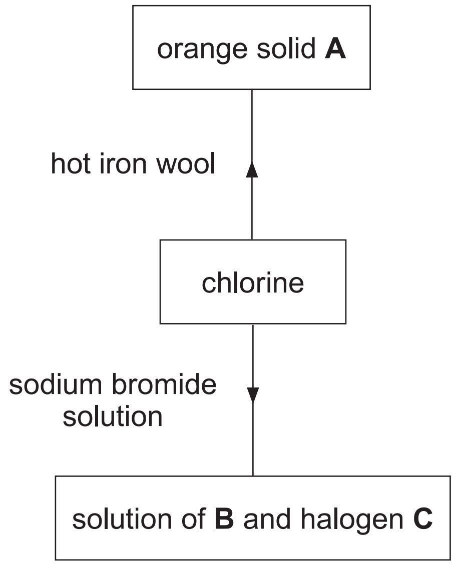
Examiner only

5. *(a)* The diagram shows some reactions of chlorine.



(i) Give the chemical **formulae** for substances **A**, **B** and **C**. [3]

A ....

B ....

C ....

(ii) State, giving a reason, the **main** safety precaution that must be taken when carrying out these reactions. [1]

....

....

(iii) State and explain how the observations for the reaction with sodium bromide would be different if the chlorine were replaced with iodine. [3]

....

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(b) Chlorine can form a number of compounds by reacting with oxygen gas. One of these is used in water treatment.

A sample of this compound contains 0.71 g of chlorine and 0.64 g of oxygen. Calculate its simplest formula. [3]

Simplest formula ....

(c) Silver nitrate solution is used to detect the presence of halide ions in solution.

(i) The equation below represents the reaction between silver nitrate solution and potassium bromide solution.

$$AgNO_3 \quad + \quad KBr \longrightarrow AgBr \quad + \quad KNO_3$$

Write the **ionic** equation for the reaction. **Include state symbols**. [2]

....

(ii) Write the balanced **symbol** equation for the reaction between calcium iodide solution and silver nitrate solution. [3]

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